TF

■ APPLIED AFFECTIVE COMPUTING

TriFusion

Trimodal Emotional Intelligence System

> A real-time affective computing platform that integrates facial expressions, speech acoustics, and natural language semantics. The system calculates statistical signal divergence using information-theoretic metrics to identify emotional masking and coordinates stateful wellness interventions.

3	23-D	KL-Div	<300ms
SENSORY MODALITIES	VECTOR INPUT DIMENSION	DIVERGENCE SCORER	CYCLE LATENCY BUDGET

PROJECT	**TriFusion** Trimodal Affect Systems Specification
TECHNICAL AREA	Multimodal Representation, Symmetric Divergence Modeling, Stateful Graphs
PRINCIPAL AUTHOR	**Lalithendra Kasula**
ACADEMIC ORG	Newton School of Technology, Rishihood University
ACADEMIC PROGRAM	B.Tech in Artificial Intelligence (2024–2028)
DEVELOPMENT SCOPE	Solo Implementation (3 fine-tuned model backbones, fusion network, state agent, telemetry, API)

TECHNICAL DESIGN REPORT & SYSTEM SPECIFICATION VERSION 1.0.0 · VISION + AUDIO + TEXT INTEGRATION

Executive Summary

Affective computing through trimodal signal fusion and stateful agentic feedback loops.

Affective intelligence systems frequently suffer from reliability constraints when analyzing isolated communication channels. Human emotion is complex and often characterized by discrepancies between verbal declarations and physiological expressions. To address this limitation, **TriFusion** integrates three distinct deep learning backbones to run continuous, parallel affect estimation across **facial expressions (vision)**, **speech acoustics (audio)**, and **semantic intent (natural language)**.

Core Contributions

The system achieves high robustness and predictive validity by establishing three major structural layers:

- **Parallel Perception Channels:** Independent fine-tuning of three neural network families—a CNN (EfficientNet-B0) optimized for spatial facial features, a self-supervised audio encoder (Wav2Vec2) for temporal vocal structures, and a text transformer (RoBERTa-base) for syntactic sentiment extraction.
- **Symmetric Information-Theoretic Telemetry:** Instead of relying on naive maximum-value voting, TriFusion computes the pairwise symmetric Kullback-Leibler (KL) divergence across the three emotion output distributions. This establishes a mathematically rigorous metric for **emotional incongruence**, signaling cognitive-physiological dissonance or emotional masking (e.g., verbalizing positive sentiment under high physiological stress).
- **Stateful Cyclic Execution Graph:** A compiled LangGraph routing architecture (WellnessAgent) continuously monitors affective state vectors and incongruence coefficients. The agent matches detected patterns to specific clinical support modules (grounding, cognitive reframing, respiratory exercises) and employs stateful counters to enforce safety gates, such as automated human-resource escalation during persistent, severe distress signatures.

> **Symmetric KL-Divergence Formulation:**
> By computing the average pairwise divergence normalized by the maximum entropy of an 8-class distribution, the system yields an incongruence index bounded in `[0.0, 1.0]`. This acts as a continuous, mathematically grounded sensor for affect misalignment.

Structural Summary

OPERATIONAL DIMENSION	TECHNICAL APPROACH	SYSTEMS OUTCOME
Modality Fusion	PyTorch MLP (Late Fusion over 23-D concatenated probabilities)	Backbone-agnostic, graceful degradation on stream loss
Divergence Engine	Average Pairwise Symmetric KL Divergence / log(8) normalization	Statistical detection of emotional masking (e.g., stress masking)
Response Loop	LangGraph state machine with memory and safety limits	Contextual, repeat-free, threshold-gated wellness feedback
Runtime Concurrency	Decoupled capture and inference threads with ring buffers	30+ FPS UI refresh rates with heavy ML backends

Problem Definition & Domain Context

The systemic vulnerabilities of unimodal affect recognition and distress-masking behaviors.

Existing emotional AI implementations are predominantly unimodal and linear. Typical installations analyze written text for sentiment or evaluate static camera frames for facial expression in isolation. These systems ignore a fundamental aspect of human behavior: **expressive dissonance**. Individuals under acute stress or social anxiety often modify their verbal output to conform to social expectations while displaying physiological indications of anxiety, anger, or sadness in their vocal delivery and micro-expressions.

The Pathology of Distress Masking

Unimodal systems are systematically blind to cognitive masking. For instance, in clinical, occupational, or customer support scenarios, a subject may verbally declare: *"Everything is fine, thank you."* A text-only transformer classifies this input as positive or neutral with high confidence.

However, the physical generation of this utterance may be accompanied by high fundamental frequency jitter (micro-tremor in vocal chords) and micro-actions of the corrugator supercilii muscle (brow furrowing). TriFusion detects this contradiction. The inability to detect these discrepancies results in critical failure modes across several application domains:

- **Digital Triage & Clinical Portals:** Crisis intervention tools rely heavily on self-reported questionnaires or explicit text inputs. This leads to high false-negative rates in detecting acute psychological distress, as individuals in crisis frequently mask their symptoms.
- **Occupational Burnout Telemetry:** In high-stress, safety-critical environments (e.g., air traffic control, emergency response centers), operator fatigue and cognitive overload manifest first in subtle acoustic features and facial micro-expressions before appearing in verbal interactions or structured reports.
- **Operational Support Systems:** Customer service bots and virtual agents fail to identify customer frustration during early stages. This delay in sentiment tracking prevents proactive service routing, resulting in low containment and high churn.

> **Systems Engineering Perspective:**
> Unimodal affect classifiers act as low-fidelity sensors with high bias. TriFusion frames affect recognition not as a simple labeling task, but as a multi-sensor alignment problem. By treating face, voice, and text as three independent indicators, the system exposes the statistical divergence between them as a primary engineering metric.

Quantitative Vulnerability Assessment

SENSOR CHANNEL	TYPICAL FAILURE MODE	IMPACT ON AFFECT MONITORING
Text Sentiment	Social desirability bias, intentional masking	False-positive "neutral/positive" classifications during high-anxiety states.
Facial Expression	Volitional masking (e.g., social smile), lighting loss	Inability to distinguish between authentic enjoyment and polite social smiling.
Speech Acoustics	Acoustic noise, flat affect confusion	False classification of high arousal (excitement) as anger due to overlapping pitch dynamics.

Technical Design Objectives

Functional and operational constraints for multi-channel affective computing pipelines.

The development of TriFusion is guided by five primary systems engineering and machine learning objectives. These objectives define the architectural choices, hardware constraints, and framework selections of the platform:

1. Real-Time Latency Budget (<300ms)

To be effective in real-time intervention scenarios, the entire loop—from physical sensory capture to modality pre-processing, concurrent model inference, neural late-fusion, information-theoretic divergence calculation, and LangGraph agent execution—must complete within a **300ms cycle budget**. This limit prevents conversational lag and ensures immediate, interactive wellness feedback.

2. Statistical Rigor in Incongruence Scoring

The measurement of emotional misalignment must not rely on ad-hoc heuristics or arbitrary classification rules. The system requires a mathematically grounded, symmetric distance metric that treats the output probability vectors of the three modalities as probability distributions over a unified emotion space. This index must scale predictably from `0.0` (complete sensor agreement) to `1.0` (maximal divergence) to allow for reliable threshold triggers.

3. Graceful Modality Degradation

In real-world environments, hardware channels are unreliable. The system must degrade gracefully when a webcam is blocked or a microphone disconnects. The late-fusion architecture must remain functional given partial inputs (e.g., text-only or vision-audio without text) by mapping missing streams to uniform probability distributions, ensuring the overall pipeline never blocks.

4. Decoupled Processing & Multi-threaded Telemetry

Executing heavy neural models (EfficientNet, Wav2Vec2, RoBERTa) sequentially on a single thread forces the I/O capture loop to block, capping display rates at <12 FPS. The architecture must decouple camera and audio capture from deep learning model execution, permitting the UI to maintain a smooth 30+ FPS refresh rate while model inference runs asynchronously in the background.

5. Safety-Gated LLM Intervention Logic

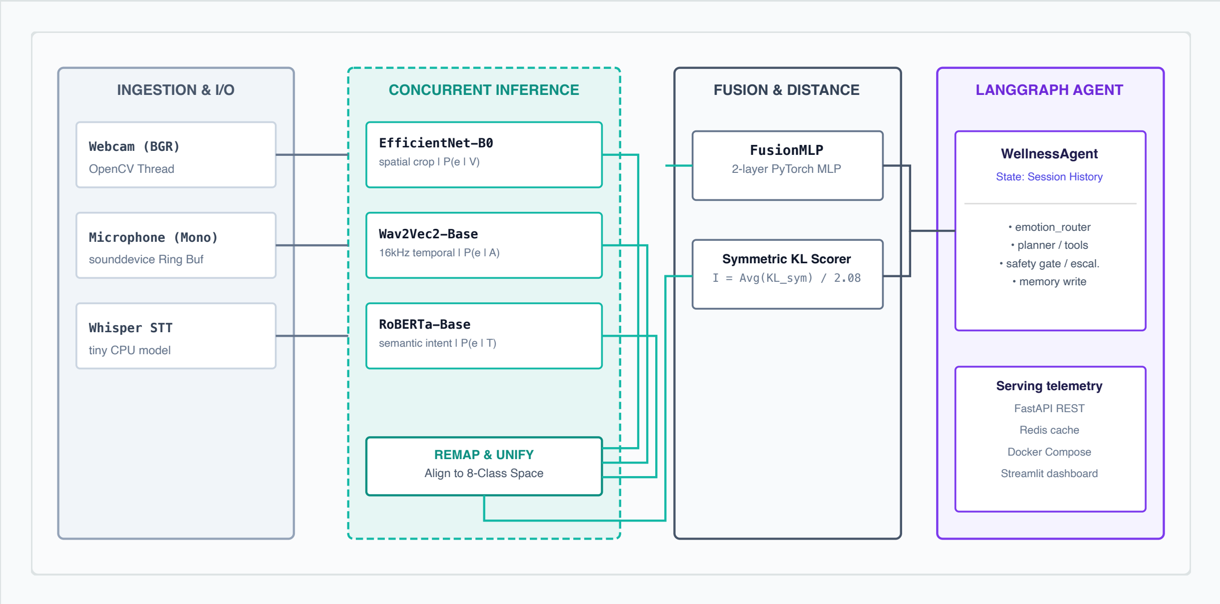
Large language models (LLMs) used for behavioral interventions must not operate in an unconstrained, state-free loop. The system requires a stateful tracking mechanism that preserves session history, limits maximum response count to prevent intervention fatigue, avoids consecutive repetitive outputs, and routes execution through strict, non-stochastic safety blocks when crisis criteria are met.

> **Performance Constraint:**
> Achieving these objectives requires isolating hardware acceleration layers (CUDA for NVIDIA, MPS for Apple Silicon, CPU fallback) and optimizing data transfers between local frontends and decoupled REST backends.

System Architecture Overview

Functional block diagram and data routing paths from sensory capture to agentic output.

The TriFusion architecture organizes data flow into four main pipeline stages: **Ingestion & Decoupled Capture**, **Parallel Feature Extraction & Inference**, **Probability Remapping & Neural Late Fusion**, and **Stateful Agentic Routing**.



The `PipelineManager` handles thread coordination. The camera capture thread runs at the native frame rate of the webcam, placing frames into a single-slot buffer (`_frame_q`). The inference thread reads from this buffer asynchronously, passing cropped faces, audio buffers, and transcripts to their respective models.

Once inference outputs are Unified, the Fusion Engine computes the incongruence coefficient. The combined state array then routes the WellnessAgent. Finally, the update is pushed to local session history (backed by Redis or an in-memory dictionary) and rendered on the Plotly-powered Streamlit layout.

Vision & Audio Perception Channels

Spatial facial expressions analysis and acoustic transformer sequence classification.

1. Spatial Vision Pipeline (Facial Expression Recognition)

The vision channel processes frames to estimate probabilities over seven facial categories. Raw BGR frames from the camera thread pass to a `FaceDetector` wrapper.

- **MediaPipe Face Detection:** Initializes MediaPipe's lightweight single-stage detector configured with `model_selection=0` (optimized for desktop distances <2 meters). If the library is missing or fails, the pipeline uses a legacy OpenCV Haar Cascade classifier fallback.
- **Proportional Context Padding:** The bounding box is expanded outward by **20% proportional padding** on all sides. This padding preserves structural context (e.g., forehead, jawline, and ear markers) which is critical for classification networks trained on standard datasets.
- **EfficientNet-B0 Backbone:** Features are extracted via an ImageNet-pretrained EfficientNet-B0. The stock single-linear head is replaced by a two-stage classification MLP: `Dropout(p=0.4) → Linear(1280, 256) → ReLU → Dropout(p=0.3) → Linear(256, 7)`.
- **Training Strategy:** Fine-tuned on the **FER2013** dataset (35,887 grayscale images) using AdamW, a learning rate of `1e-4`, and weight decay of `0.01`. The model employs weighted cross-entropy loss with `0.1` label smoothing to handle the high noise floor (~20%) of FER2013 labels.

2. Acoustic Audio Pipeline (Speech Emotion Recognition)

The audio channel isolates physical vocal properties (intonation, frequency shifts, energy) from semantic content.

- **Microphone Circular Buffer:** Audio is captured continuously at **16 kHz mono** in 0.1-second blocks using a background `sounddevice.InputStream` thread, accumulating into a 6-second circular queue. On validation cycles, the most recent **3-second sliding window** (48,000 samples) is retrieved.
- **Wav2Vec2 Feature Extraction:** The window is processed using HuggingFace's `Wav2Vec2Processor`, which normalizes the waveform to zero-mean and unit-variance.
- **Wav2Vec2-Base Model:** The backbone uses the **facebook/wav2vec2-base** checkpoint (95M parameters). Features are mean-pooled across the time axis before routing to the classification layer.
- **Staged Fine-Tuning:** Trained on the **RAVDESS** dataset. To prevent gradient explosion in the initial stages, the convolutional feature encoder is frozen for the first epoch (Phase 1). The encoder is then unfrozen for a full end-to-end fine-tuning over 15 epochs at a learning rate of `3e-5` (Phase 2), controlled by an early-stopping monitor.

> **Modality Remapping Constraint:**
> FER2013 lacks a native "calm" class. During alignment mapping, the vision probability for `calm` is set to `0.0` and normalized. If the user presents a calm state, the audio and text modalities pull the fused probability distribution toward "calm" in the late-fusion MLP.

Natural Language & Real-Time Transcription

Whisper speech-to-text decoding and robust semantic affect classification.

1. Real-Time Speech-to-Text Pipeline

The textual pipeline extracts semantic meaning from spoken words. To support real-time deployment without requiring heavy GPU memory clusters, the transcription is handled by a local CPU-optimized Whisper decoder.

- **Whisper-Tiny Model:** The system deploys OpenAI's **Whisper-tiny** model (~39M parameters). This model achieves a low processing latency of **~150ms per 3-second audio block** on standard x86 and Apple Silicon CPUs, meeting the requirements of the real-time loop.
- **Decoding Optimizations:** To minimize language detection overhead, the language is locked to English (`language="en"`). The decoder runs with `fp16=False` to ensure numerical stability on CPU fallbacks and disables `condition_on_previous_text` to prevent the carries of previous hallucinations over empty audio frames.

2. Semantic Affect Classification Pipeline

Once transcription yields a text segment, it is processed for semantic affect classification.

- **RoBERTa-Base Model:** Semantic intent is classified using the **roberta-base** transformer configuration (125M parameters), which replaces BERT's static masking with dynamic masking and drops Next Sentence Prediction (NSP) training.
- **GoEmotions Remapping:** The model is fine-tuned on the **GoEmotions** corpus (58,009 Reddit comments). The original 27 fine-grained emotion labels are collapsed into TriFusion's unified 8-class emotional schema at the dataset level, allowing the model head to output 8 logits directly.
- **Hyperparameters:** The classifier is trained over 4 epochs using a learning rate of `2e-5`, weight decay of `0.01`, and a linear warmup ratio of `0.1`. Tokenization uses `RobertaTokenizerFast` capped at a maximum length of 128 tokens.
- **Null Input Safeguard:** Natural conversation contains pauses and noise. Text segments shorter than **2 characters** bypass the transformer entirely and return a uniform distribution (`P(e) = 0.125` for each class). This prevents the model from outputting confident, biased classifications on silent frames.

> **Transcription Uncertainty:**
> Whisper-tiny has a word error rate (WER) of ~10% on casual, informal speech. By converting output logits to a smooth probability distribution rather than hard labels, the Fusion MLP treats text classifications as soft, probabilistic distributions, which reduces the impact of transcription errors.

Modality Configuration Reference

MODEL BACKBONE	INPUT PARAMETERS	TARGET DATASET	OUTPUT SPACE
EfficientNet-B0	224×224 RGB image	FER2013	7-Class Probability Vector

Neural Late Fusion Framework

Modality integration via a two-layer Multi-Layer Perceptron.

A key architectural decision in TriFusion is the use of **late fusion** over early or intermediate fusion. Early fusion concatenates raw features (e.g., image pixels, raw audio waves) at the input layer, which requires training high-dimensional, unstable cross-modal layers. Intermediate fusion joins hidden-state embeddings, which introduces high computational costs and requires joint backpropagation through three large backbone models.

In contrast, late fusion combines the normalized class probability outputs of each modality model. This approach is highly efficient, makes the overall architecture backbone-agnostic, and allows individual sensor models to be updated or retrained independently.

FusionMLP Architecture

The late-fusion coordinator is implemented as a PyTorch Multi-Layer Perceptron (MLP) mapping a **23-dimensional concatenated input vector** (vision: 7, audio: 8, text: 8) to the final unified 8-class emotional space.

```
# src/fusion/fusion_model.py
class FusionMLP(nn.Module):
    def __init__(self, input_dim=23, output_dim=8):
        super().__init__()
```

The Role of Batch Normalization

Modality probability vectors are scale-mismatched before concatenation. Because FER2013 is a 7-class system and RAVDESS/GoEmotions use 8 classes, the raw values of the input blocks span different variance ranges. Placing a `BatchNorm1d` layer immediately after the first linear projection normalizes feature scale distributions, stabilizing gradient updates during training.

Synthetic Training Procedure

Because real-world datasets with simultaneous, synchronized face, voice, and text annotations are rare, the FusionMLP is trained on a synthetic dataset of 8,000 samples designed to match real-world distributions. The training dataset consists of:

- **Congruent Samples (70%):** All three modalities point to the same ground-truth emotion. The model learns to aggregate agreeing inputs and output a confident classification.
- **Incongruent Samples (30%):** Modalities are assigned different emotions. In these cases, the audio model (RAVDESS) is configured as the most reliable indicator, training the network to prioritize physiological voice cues over verbal declarations.

Input probabilities are generated using Dirichlet distributions with parameter `α = 0.3`, which creates realistic, skewed class spreads rather than flat distributions.

Information-Theoretic Incongruence Scoring

Measuring affect alignment using symmetric Kullback-Leibler divergence.

Rather than relying on heuristics, TriFusion uses information theory to quantify the disagreement between different input channels. The system treats the unified 8-class probability vectors from the vision (P_V), audio (P_A), and text (P_T) channels as independent probability distributions over the same emotional event space.

Symmetric KL Divergence

The standard Kullback-Leibler (KL) divergence measures the informational difference between two distributions, but it is asymmetric: $D_{KL}(P || Q) \neq D_{KL}(Q || P)$. This asymmetry is problematic for sensor fusion, where we want to measure the mutual disagreement between two channels without prioritizing one over the other.

To address this, TriFusion computes the **symmetric KL divergence**, which averages the divergence in both directions:

$$D_{sym}(P \; || \; Q) = [\; D_{KL}(P \; || \; Q) + D_{KL}(Q \; || \; P) \;] \; / \; 2.0$$

The system evaluates the symmetric divergence across all three modality pairs and computes the average:

$$\text{Average KL} = [\; D_{sym}(P_V \; || \; P_A) + D_{sym}(P_V \; || \; P_T) + D_{sym}(P_A \; || \; P_T) \;] \; / \; 3.0$$

Normalization & Clamping

While raw KL divergence is theoretically unbounded, the maximum possible symmetric divergence for two discrete 8-class distributions occurs when one distribution is fully concentrated on a single class ($p_i = 1$) and the other is uniform ($q_i = 0.125$). This maximum value is $\ln(8) \approx 2.08$ nats. By dividing the average KL by 2.08 and clamping the result to `[0.0, 1.0]`, the system produces a normalized incongruence coefficient.

Operational Thresholds

The normalized incongruence score is mapped to three operational regions:

0.0 – 0.3 (Aligned) `Aligned`

0.3 – 0.7 (Moderate) `Moderate`

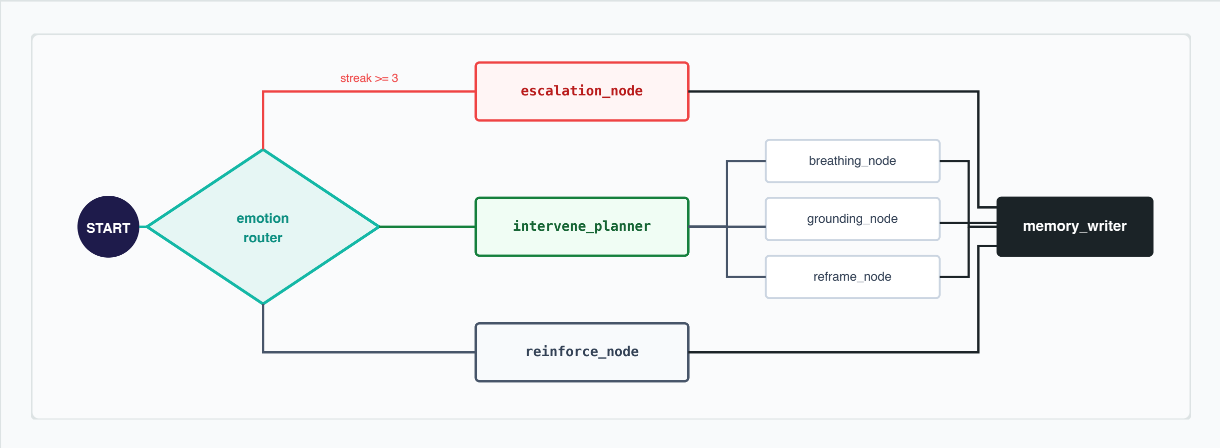
0.7 – 1.0 (Incongruent) `Masking`

Mathematical Stability Guard:
To prevent division-by-zero errors during KL divergence calculations when a class probability is 0, a small epsilon value (`ε = 1e-8`) is added to all elements. The vector is then re-normalized before calculation: `arr = (arr + 1e-8) / sum(arr + 1e-8)`.

LangGraph Stateful Conversation Orchestration

WellnessAgent state machine topology, tool nodes, and confidence routing logic.

TriFusion uses a stateful, cyclical orchestration graph built with LangGraph to coordinate wellness responses. Rather than processing frames in isolation, the system threads an `AgentState` schema containing the full session history across execution nodes.



State Routing Nodes & Execution Flow

- **emotion_router:** Acts as the entry node. It evaluates the current frame's dominant emotion and incongruence score. If the incongruence exceeds `0.7` for **3 consecutive frames** during high-severity emotional states (fear/sadness/anger), it routes directly to the `escalation` node. If incongruence is high but the streak is shorter, or if emotion severity is high, it routes to the `intervention_planner`. If the state is happy or calm, it transitions to `reinforce`. Otherwise, it routes to the default `affirmation` block.
- **intervention_planner:** Prevents repetitive interventions by checking the previous action stored in the state history. If the default intervention tool matches the last-used tool, the planner rotates to an alternative technique, maintaining variety in conversational responses.
- **Leaf Tool Nodes:** Call the Groq API (LLaMA-3.3-70B-versatile) to dynamically generate tailored responses based on structured system prompts. The `cognitive_reframe` node passes the raw transcribed text from Whisper so the model can reframe the user's specific statements.
- **escalation_node:** Unlike other tools, this node does not call the LLM. To ensure reliability during crises, it immediately outputs a pre-written support resource text block and locks the state (`escalated = True`) to prevent further LLM calls.
- **memory_writer:** The final node in the graph, appending the processed frame to the session history log.

Technology Stack & Architectural Decisions

Evaluation and rationale for core technologies, frameworks, and storage layers.

The TriFusion stack is selected to balance pipeline execution speeds, deployment reproducibility, and information-theoretic modeling requirements:

Systems Infrastructure & Serving

- **FastAPI API Core:** Exposes the inference and agentic pipelines via high-performance ASGI endpoints. This separation allows the text inference and LangGraph orchestration layers to scale independently of the hardware-linked Streamlit visualization layer, enabling integrations with external systems like mobile apps or backend analytics pipelines.
- **Redis Session Persistence:** Used as an in-memory database to store serialized `EmotionFrame` sequences. Storing history logs in Redis allows the dashboard to query session analytics without maintaining state in the Streamlit application memory. If Redis is offline, the system falls back to an in-memory dictionary.
- **Docker Containerization:** Standardizes development environments across different systems using a multi-stage build. The services are split into decoupled containers coordinated by `docker-compose`:

```
# docker-compose.yml (summary)
services:
  api:
    build: .
    command: python src/api/main.py
    ports: ["8000:8000"]
  dashboard:
    build: .
    command: streamlit run dashboard/app.py
    ports: ["9002:9002"]
  redis:
    image: redis:alpine
```

Framework Rationale

FRAMEWORK	DIRECT COMPETITOR	ARCHITECTURAL RATIONALE
LangGraph	Sequential LangChain / LlamaIndex Agent	Enforces cyclical state transitions (e.g., routing back to memory) and custom edge conditions better than standard linear graphs.
Streamlit	Custom React / Vue Frontend	Provides rapid UI prototyping and easy integration with local webcam and audio hardware drivers.
FastAPI	Flask / Django	Native support for async request parsing, automatic Pydantic validation, and lower request latencies.
Redis	SQLite / PostgreSQL	Provides sub-millisecond read/write speeds, key TTL expirations for security, and holds temporary session states.

Performance Telemetry & Concurrency

Analysis of the multi-threaded ingestion model and hardware latency evaluations.

The 12 FPS Bottleneck

In naive single-threaded affect processing systems, I/O frame capture and model inference run sequentially. Under this model, each iteration blocks for the duration of both tasks (33ms capture + 50ms model processing), which limits rendering rates to <12 FPS and causes significant visual lag.

Decoupled Threading Architecture

To maintain a responsive interface, TriFusion implements a decoupled threading model managed by `PipelineManager`:

- **CaptureThread:** Runs at **30+ FPS**, performing only raw frame ingestion via OpenCV and writing to a single-slot queue (`_frame_q`). It bypasses all ML processing to prevent thread blocking.
- **InferenceThread:** Consumes frames from the queue asynchronously, executing the three pipeline models and writing the results to a shared data structure protected by a threading lock (`_result_lock`).
- **Telemetry Refresh:** The Streamlit UI displays the latest frame from the capture buffer at 30 FPS, while rendering the bounding box from the latest completed inference cycle. Streamlit's `st.fragment` decorator is used to refresh the telemetry charts and indicators independently, preventing full-page re-renders.

Hardware Latency Profile

The table below details loop execution latencies across different hardware environments. The system automatically detects and utilizes hardware acceleration (using PyTorch's CUDA and MPS backends), falling back to standard CPU execution if no GPU is present.

HARDWARE SETUP	INFERENCE SPEED	AVERAGE LOOP LATENCY	OPERATIONAL STATUS
NVIDIA RTX 4090 (CUDA)	50 – 60 FPS	~18 ms	Real-time, zero-lag rendering
Apple M3 Pro (MPS)	25 – 35 FPS	~35 ms	Real-time, smooth rendering
Apple M1 (MPS)	15 – 20 FPS	~60 ms	Usable, minor latency
Intel i7 CPU (Fallback)	5 – 10 FPS	~140 ms	Low frame rate, visible latency

Runtime Latency Breakdown:
Whisper transcription dominates latency, taking ~150ms on standard CPUs. By limiting textual analysis updates to every 5th inference frame, the system distributes processing loads and keeps the average cycle latency below the 300ms budget limit.

Engineering Challenges & Mitigations

Handling missing sensor inputs, class mismatch, and overconfident models.

Challenge 1: Input Failures & Graceful Degradation

In real-world applications, physical camera or microphone feeds frequently drop out, disconnect, or return corrupt data.

Mitigation: The system implements a robust error handling structure. If the camera thread fails to capture a face (e.g., due to low light or occlusion), the `VisionInference` module returns a uniform probability distribution ($P(e_i) = 0.125$ for all classes). This uniform representation indicates "maximum uncertainty" to the late-fusion MLP, meaning the final classification is determined by the remaining active modalities without blocking the pipeline.

Challenge 2: Class Space Misalignment

The datasets used to train individual models use different emotion classification spaces: FER2013 uses 7 classes, RAVDESS uses 8 classes, and GoEmotions uses 27 classes. These differences prevent direct vector comparison.

Mitigation: TriFusion maps all modality outputs onto a unified 8-class emotional schema: `neutral, happy, sad, angry, fearful, surprised, disgusted, calm`. Mapping functions in `src/utils/emotion_mapper.py` group GoEmotions labels (e.g., remapping *grief* and *remorse* to *sad*) and handle FER2013's missing "calm" class by setting its initial probability to 0 before final vector normalization.

Challenge 3: Overfitting and Calibration in Transformers

Fine-tuning deep networks on small emotion datasets (like RAVDESS) often leads to overconfident probability predictions, which distorts the KL divergence calculations.

Mitigation: The system applies several regularization techniques during model training:

- **Label Smoothing (0.1):** Applied to cross-entropy loss calculations during EfficientNet-B0 and Wav2Vec2 training to prevent the models from outputting extreme probabilities.
- **Cosine Annealing LR:** Smoothes learning rate transitions to prevent training instability.
- **Validation F1 Checkpointing:** Saves checkpoints based on validation-set weighted F1 scores rather than training loss, preventing overfitting to training samples.

Challenge 4: LLM Repetition and Intervention Fatigue

Under persistent negative emotional states, the LangGraph agent may repeatedly trigger the same intervention node, leading to repetitive and unhelpful suggestions.

Mitigation: The agent implements clinical rotation logic in the `intervention_planner`. If the preferred intervention tool matches the one used in the previous frame, the system dynamically swaps it for an alternative, deterministically chosen technique. Additionally, a session limit capped at **5 interventions** prevents conversational overload.

Future Work & Technical References

Potential research areas, optimizations, and academic literature index.

Future Work & Optimizations

- **Region-of-Interest Micro-Expression Tracking:** Integrating facial landmark tracking to isolate active facial sub-regions (e.g., mouth, eyes, eyebrows) and capture brief, involuntary micro-expressions that indicate hidden stress more reliably than general classifications.
- **Cross-Encoder Calibration:** Implementing Temperature Scaling or Platt Scaling on individual modality outputs. Calibrating modality probabilities ensures that confidence scores correspond to actual predictive accuracies, which improves the precision of the KL divergence scorer.
- **Decoupled Multi-Subject Support:** Extending the `PipelineManager` and MediaPipe tracking threads to detect and track multiple faces and voices simultaneously, allowing the system to run multi-user group telemetry.

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